

Mathematical and Computational Methods

Exercise Sheet 9: Integration I: the definite integral and substitution

Tutorial route (►, in this order, about 80 minutes): 1.1(a), 2.2(a–c), 2.3, 2.4, 2.5, 2.7, 2.9. Everything else is for home study.

Levels: ★ drill, ★★ core, ★★★ challenge. Give exact answers unless a decimal is asked for.

1 Warm-up drills

Riemann sums and signed area

Exercise 1.1 (► ★ drill — a Riemann sum and signed area). (a) Estimate the area under $f(x) = x^2$ on $[0, 2]$ using $n = 4$ rectangles of equal width and the height at the *right* end of each strip. Sketch the curve with the four rectangles, and say whether your estimate is an over- or under-estimate, and why.

(b) Sketch $g(x) = x - 1$ on $[0, 2]$ and shade the region between the graph and the x -axis, marking the part that counts as *negative* signed area. Find the net signed area $\int_0^2 (x - 1) dx$ and the total (unsigned) area.

Indefinite integration: the power rule and the table

Exercise 1.2 (★ drill — rewrite first, then integrate). Write each integrand as a sum of powers of x and standard forms, then integrate:

$$(a) \int \left(\sqrt{x} + \frac{1}{x^2} \right) dx, \quad (b) \int \left(2e^x - \frac{3}{x} + 5 \cos x \right) dx, \quad (c) \int \frac{x^2 - 3x + 2}{x} dx.$$

Standard integrals (including arctan/arcsin)

Exercise 1.3 (★ drill — standard forms with a linear inside). Evaluate the indefinite integrals

$$(a) \int e^{3x} dx, \quad (b) \int \sec^2(2x) dx, \quad (c) \int \frac{dx}{9 + x^2}, \quad (d) \int \frac{dx}{\sqrt{4 - x^2}}, \quad (e) \int \cos(5x) dx.$$

FTC Part 2: evaluating definite integrals

Exercise 1.4 (★ drill — definite integrals by FTC 2). Evaluate

$$(a) \int_1^3 x^2 dx, \quad (b) \int_0^{\pi/2} \cos x dx, \quad (c) \int_1^e \frac{dx}{x}, \quad (d) \int_0^1 e^{2x} dx.$$

FTC Part 1: differentiating an integral

Exercise 1.5 (★ drill — FTC Part 1). Differentiate with respect to x (do *not* try to evaluate the integral):

$$(a) \frac{d}{dx} \int_0^x e^{-t^2} dt, \quad (b) \frac{d}{dx} \int_2^x \sin(t^3) dt, \quad (c) \frac{d}{dx} \int_1^x \frac{dt}{1 + t^4}.$$

Substitution (reverse chain rule and f'/f)

Exercise 1.6 (★ drill — substitution and the f'/f pattern). Each integrand contains an “inside” function whose derivative is present. Integrate by substitution; in (c) and (d) the numerator is the derivative of the denominator.

$$(a) \int 2x(x^2 + 1)^5 dx, \quad (b) \int \cos x e^{\sin x} dx, \quad (c) \int \frac{2x}{x^2 + 5} dx, \quad (d) \int \frac{e^x}{e^x + 1} dx.$$

Exercise 1.7 (★ drill — changing the limits). Evaluate $\int_1^2 x e^{x^2} dx$ by changing the limits to the new variable.

No elementary antiderivative

Exercise 1.8 (★ drill — elementary antiderivatives?). For each function, state whether it has an elementary antiderivative.

$$(a) e^{-x^2}, \quad (b) x e^{-x^2}, \quad (c) \frac{\sin x}{x}, \quad (d) \sin(x^2), \quad (e) x \cos(x^2).$$

2 Tutorial problems

Exercise 2.1 (★★ core — Riemann sums against the exact area). Let $f(x) = 1 + x^2$ on $[0, 3]$, cut into $n = 3$ strips of width 1.

- Compute the right-endpoint and left-endpoint Riemann sums.
- Using FTC Part 2, find the exact area and check it lies between your two estimates.
- Now use n strips of width $\Delta x = \frac{3}{n}$ and right endpoints $x_k = \frac{3k}{n}$. Show that the right-endpoint sum is $S_n = 3 + \frac{9(n+1)(2n+1)}{2n^2}$, check that $n = 3$ gives your answer to (a), and find $\lim_{n \rightarrow \infty} S_n$. (You may quote $\sum_{k=1}^n k^2 = \frac{1}{6}n(n+1)(2n+1)$.)

Exercise 2.2 (► ★★ core — FTC evaluation). (FTC evaluation — worked-example anchor.)

- Evaluate $\int_1^4 x^2 dx$ and state in one sentence what the number represents geometrically.
- Evaluate $\int_0^1 \left(3x^2 + \frac{4}{1+x^2}\right) dx$.
- Find the positive value of a for which $\int_0^a (2x + 1) dx = 12$.

Exercise 2.3 (► ★★ core — displacement and distance (physics)). [Physics / applied] A particle moves on a line with velocity $v(t) = 3t^2 - 12t + 9$ (m s⁻¹) for $0 \leq t \leq 3$.

- Find the net displacement $\int_0^3 v dt$.
- Find the total distance travelled, $\int_0^3 |v| dt$.

Exercise 2.4 (► ★★ core — a probability density (data science)). [Data science / probability] A continuous random variable X has probability density $f(x) = kx(1-x)$ on $[0, 1]$ and $f(x) = 0$ elsewhere. A density must satisfy $f(x) \geq 0$ and $\int_0^1 f = 1$, and $\mathbb{P}(X \leq c) = \int_0^c f$ (probability is area).

- (a) Find k , and check that $f \geq 0$.
 (b) Find $\mathbb{P}\left(X \leq \frac{1}{2}\right)$ and explain briefly why this value is to be expected.

Exercise 2.5 (► ★★ core — *stating the FTC; FTC 1 with the chain rule*). [Mathematics / pure]

- (a) State both parts of the Fundamental Theorem of Calculus, including the hypothesis on f .
 (b) Let $g(x) = \int_0^{x^2} \cos t \, dt$. Find $g'(x)$ in two different ways and check they agree.

Exercise 2.6 (★★ core — *spot the error*). (Spot the error / conceptual.)

- (a) A student writes $\int_{-1}^1 \frac{dx}{x^2} = \left[-\frac{1}{x}\right]_{-1}^1 = (-1) - (1) = -2$. The integrand is positive everywhere it is defined, so a negative “area” is impossible. Which theorem did the student use? Check its hypothesis and explain what went wrong.
 (b) True or false: “ $\int_a^b f(x) \, dx$ is a function of x .” Justify.

Exercise 2.7 (► ★★ core — *substitution with a limit change*). (Substitution with a limit change — worked-example anchor.) Evaluate $\int_0^2 x\sqrt{x^2+1} \, dx$ by converting the limits to the new variable.

Exercise 2.8 (★★ core — *a stray x to rewrite*). Evaluate $\int_0^3 x\sqrt{x+1} \, dx$. (Here the substitution leaves a stray x behind — rewrite it in terms of u .)

Exercise 2.9 (► ★★ core — *two routes to a logarithm*). Evaluate two integrals, contrasting the methods:

$$(a) \text{ by the } f'/f \text{ pattern, } \int \tan x \, dx, \quad (b) \text{ by substitution, } \int_1^2 \frac{\ln x}{x} \, dx.$$

3 Further practice (home)

Properties of the definite integral

Exercise 3.1 (★ drill — *properties of the definite integral*). Suppose $\int_0^2 f = 5$ and $\int_2^5 f = -2$. Using only the properties of the definite integral (do not find f), evaluate

$$(a) \int_0^5 f, \quad (b) \int_5^2 f, \quad (c) \int_2^2 f, \quad (d) \int_0^2 3f, \quad (e) \int_5^0 f.$$

Substitution — indefinite and definite

Exercise 3.2 (★ drill — *substitution drill*). Integrate

$$(a) \int x^2(x^3+1)^7 \, dx, \quad (b) \int \sin^4 x \cos x \, dx, \quad (c) \int \frac{(\ln x)^2}{x} \, dx, \quad (d) \int x e^{-x^2} \, dx.$$

Exercise 3.3 (★★ core — *definite substitution*). Evaluate by substitution. In (a) and (c) change the limits; do (b) by *both* routes (return to x , then change the limits) and check that they agree.

$$(a) \int_0^{\pi/2} \sin x \cos x \, dx, \quad (b) \int_0^1 \frac{x}{(1+x^2)^2} \, dx, \quad (c) \int_0^1 \frac{2x}{x^2+1} \, dx.$$

arctan and arcsin targets**Exercise 3.4** (★★ core — *inverse-trig targets and completing the square*). Evaluate

$$(a) \int_0^{\sqrt{3}} \frac{dx}{1+x^2}, \quad (b) \int_0^1 \frac{dx}{\sqrt{4-x^2}}, \quad (c) \int_0^2 \frac{dx}{4+x^2}, \quad (d) \int \frac{dx}{x^2+2x+2}.$$

In (d), complete the square first.

Signed area versus total area**Exercise 3.5** (★★ core — *signed area and total area*). For each function find both the net signed area and the total (painted) area over the stated interval.

(a) $\sin x$ on $[0, 2\pi]$.

(b) $x^2 - 1$ on $[0, 2]$.

Symmetry**Exercise 3.6** (★★ core — *odd and even on a symmetric interval (maths)*). [Mathematics / pure] Let $a > 0$ and let f be continuous on $[-a, a]$. Using the substitution $u = -x$ on $\int_{-a}^0 f$, prove that $\int_{-a}^a f(x) dx = 0$ whenever f is odd, and that it equals $2 \int_0^a f(x) dx$ whenever f is even.**Challenges and extensions****Exercise 3.7** (★★★ challenge — *FTC 1 with the chain rule*). Use FTC Part 1 together with the chain rule to differentiate

$$(a) \frac{d}{dx} \int_0^{x^2} e^t dt, \quad (b) \frac{d}{dx} \int_x^{x^2} \ln(1+t^2) dt.$$

Exercise 3.8 (★★★ challenge — *completing the square, log plus arctan*). A linear numerator over an irreducible quadratic splits into a logarithm plus an arctangent. Evaluate $\int \frac{x+1}{x^2+4x+5} dx$. (Hint: complete the square below, and write the numerator as a multiple of the denominator's derivative plus a constant.)**Exercise 3.9** (★★★ challenge — *no elementary antiderivative (data science)*). [Data science / numerical] Consider $I = \int_0^1 e^{-x^2} dx$ and $J = \int_0^1 x e^{-x^2} dx$.

(a) Exactly one of them can be evaluated by FTC 2 with an elementary antiderivative. Which one? Evaluate it, and explain why FTC 2 cannot evaluate the other in elementary terms.

(b) Bracket the other integral using left- and right-endpoint Riemann sums with $n = 2$ ($\Delta x = \frac{1}{2}$). Its true value is ≈ 0.7468 ; is it inside your bracket?